

# Andrew Zhao

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## POSITIONS

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**Sandia National Laboratories** (Livermore, CA) 2026–present  
*Senior Member of Technical Staff*

**Sandia National Laboratories** (Livermore, CA) 2024–2026  
*Gil Herrera Postdoctoral Fellow in Quantum Information Science*

## EDUCATION

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**PhD in Physics, University of New Mexico** 2018–2024  
Advisor: Akimasa Miyake  
Dissertation: *Learning, Optimizing, and Simulating Fermions with Quantum Computers*  
Chairman’s Award for Best Dissertation, Department of Physics and Astronomy, 2024

**BSc in Physics, University of Maryland, College Park** 2013–2018

## RESEARCH EXPERIENCE

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**Google Quantum AI** (San Francisco, CA) 2022  
*Student Researcher*  
Host: Nicholas Rubin

**Joint Center for Quantum Information and Computer Science** (University of Maryland, College Park) 2016–2017  
*Undergraduate Research Assistant*  
Mentor: Shelby Kimmel

## GRANTS

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**“Investigating quantum algorithms and complexity for applications to physical simulation”** 2024–2026  
US\$236,000  
Project PI. Funded by the Laboratory Directed R&D program at Sandia National Laboratories, Computing & Information Sciences Division, as part of the Gil Herrera Fellowship. Supported self (postdoc) and two graduate student interns.

## PUBLICATIONS

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- E. J. Bobrow, R. W. Chien, A. Kononov, J. S. Nelson, **A. Zhao**, L. K. Kovalsky, A. D. Baczewski, “Thermal state preparation for quantum simulations of warm dense matter,” *in preparation*, 2026.
- A. Christensen, **A. Zhao**, “Learning fermionic linear optics with Heisenberg scaling and physical operations,” [arXiv:2602.05058](https://arxiv.org/abs/2602.05058), 2026.
- B. Harrison, V. Iyer, O. Parekh, K. Thompson, **A. Zhao**, “Fermionic Insights into Measurement-Based Quantum Computation: Circle Graph States Are Not Universal Resources,” [arXiv:2510.05557](https://arxiv.org/abs/2510.05557), 2025. *Accepted at TQC 2026, non-proceedings track.*
- A. Zhao**, “Learning the Structure of Any Hamiltonian from Minimal Assumptions,” *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*, 1201, 2025. *Also accepted at QIP 2025.*
- A. Zhao**, N. C. Rubin, “Expanding the reach of quantum optimization with fermionic embeddings,” *Quantum* **8**, 1451, 2024.
- A. Zhao**, A. Miyake, “Group-theoretic error mitigation enabled by classical shadows and symmetries,” *npj Quantum Information* **10**, 57, 2024.
- R. Babbush, W. J. Huggins, D. W. Berry, S. F. Ung, **A. Zhao**, D. R. Reichman, H. Neven, A. D. Baczewski, J. Lee, “Quantum simulation of exact electron dynamics can be more efficient than classical mean-field methods,” *Nature Communications* **14**, 4058, 2023.
- A. K. Daniel, Y. Zhu, C. H. Alderete, V. Buchemavari, A. M. Green, N. H. Nguyen, T. G. Thurtell, **A. Zhao**, N. M. Linke, A. Miyake, “Quantum computational advantage attested by nonlocal games with the cyclic cluster state,” *Physical Review Research* **4**, 033068, 2022.
- A. Zhao**, N. C. Rubin, A. Miyake, “Fermionic Partial Tomography via Classical Shadows,” *Physical Review Letters* **127**, 110504, 2021.
- A. Zhao**, A. Tranter, W. M. Kirby, S. F. Ung, A. Miyake, P. J. Love, “Measurement reduction in variational quantum algorithms,” *Physical Review A* **101**, 062322, 2020.

## Patents

- N. C. Rubin, **A. Zhao**, “Solving quadratic optimization problems over orthogonal groups using a quantum computer,” [US Patent US 2024 0296367 A1](#), 2024.

## TALKS

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### Invited

- “Learning the structure of any Hamiltonian from minimal assumptions” Feb 2026  
Visa Research Quantum Seminar
- “Quantum learning theory as a tool for faster quantum simulation” Feb 2026  
Sandia Quantum Information Science Seminar
- “Hamiltonian learning: A survey of recent progress” Mar 2025  
University of Sherbrooke Institut quantique Colloquium
- “Group-theoretic error mitigation enabled by classical shadows and symmetries” Nov 2023  
University of Oxford Quantum Technology Theory Seminar
- “Quantum relaxation for quadratic programs over orthogonal matrices” Oct 2023  
INFORMS Annual Meeting
- “Efficient and robust learning of quantum many-body properties with classical shadows” Jun 2023  
Sandia Quantum Computer Science Seminar
- “Efficient and robust learning of fermionic reduced density matrices with classical shadows” Jan 2023  
UC Berkeley Quantum Many-Body Seminar

### Contributed

- “Learning the structure of any Hamiltonian from minimal assumptions” Jun 2025  
ACM Symposium on Theory of Computing (STOC)
- “Learning the structure of any Hamiltonian from minimal assumptions” Mar 2025  
APS Global Physics Summit
- “Learning the structure of any Hamiltonian from minimal assumptions” Feb 2025  
Quantum Information Processing (QIP) Conference
- “Nearly Heisenberg-limited Hamiltonian learning with few assumptions” Oct 2024  
Southwest Quantum Information and Technology (SQuInT) Workshop
- “Group-theoretic error mitigation enabled by classical shadows and symmetries” Oct 2023  
Southwest Quantum Information and Technology (SQuInT) Workshop
- “Calibration-free quantum error mitigation with classical shadows” Mar 2023  
APS March Meeting
- “Fermionic partial tomography via classical shadows” Mar 2021  
APS March Meeting
- “Measurement reduction in variational quantum algorithms” Mar 2020  
APS March Meeting

## TEACHING

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### Courses

- General Physics (PHYC 151), Graduate Supplemental Instructor Spring 2019  
University of New Mexico, Department of Physics and Astronomy
- General Physics Laboratory (PHYC 151L), Graduate Lab Instructor Fall 2018  
University of New Mexico, Department of Physics and Astronomy

### Guest Lectures

- “Classical shadows are both simpler and more complicated than you think” Mar 2025  
University of Sherbrooke, Institut quantique, seminar
- “Quantum metrology and distributed quantum sensing” Jun 2023  
University of New Mexico, Center for Quantum Information and Control, summer course
- “Random unitary evolution,  $t$ -designs, and applications to quantum chaos” Jun 2021  
University of New Mexico, Center for Quantum Information and Control, summer course
- “Continuous symmetries and  $O(N)$  models” Jun 2020  
University of New Mexico, Center for Quantum Information and Control, summer course
- “Simple Lie groups and simple Lie algebras” Jun 2019  
University of New Mexico, Center for Quantum Information and Control, summer course

## ADVISING

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### Sandia Graduate Interns

- Ramya Bhaskar, *University of Washington, PhD student* 2024–2025  
– Now a postdoc at Sandia
- Aria Christensen, *Ohio State University, PhD student* 2024–2026

## PROFESSIONAL SERVICE

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### Journal Referee

- Physical Review Letters • Physical Review A • Physical Review Research • PRX Quantum • Journal of Chemical Theory and Computation • Nature Communications • npj Quantum Information • Quantum • Communications in Mathematical Physics

### Conference Referee

- QIP • TQC • QSim • QCTiP

### Grant Proposal Reviewer

- Sandia LDRD program